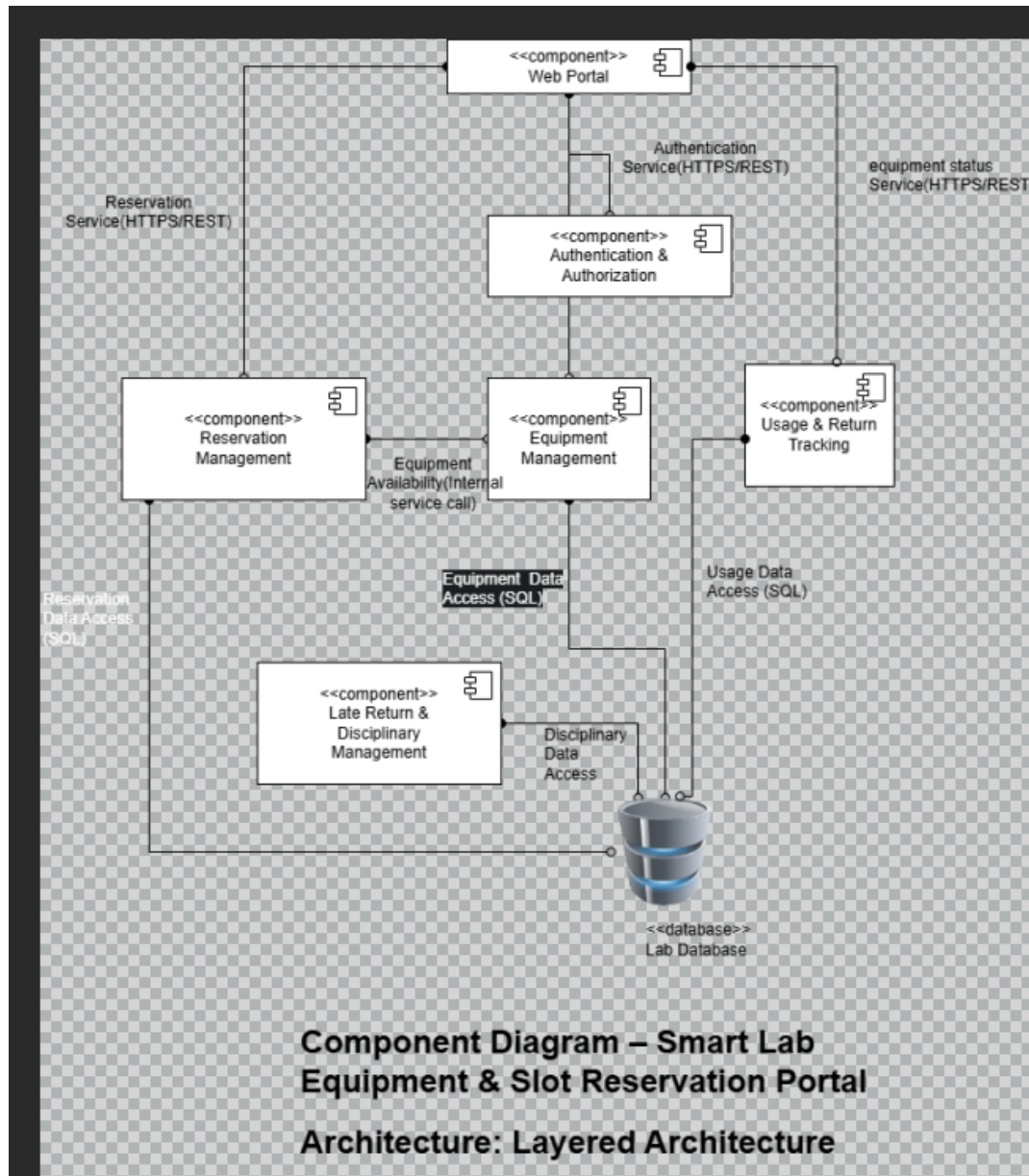


Architectural Pattern Justification

Smart Lab Equipment & Slot Reservation Portal

Selected Architecture: Layered Architecture



We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal because it offers separation of responsibilities and fits well with the different functions of the portal. The architecture divides the system into layers: Web Portal, Authentication & Authorization Reservation Management, Equipment Management, Usage & Return Tracking Late Return & Disciplinary Management and Lab Database.

Reasons for Selection

1. Separation of Responsibilities:

Each part of the system has an focused job. The Reservation Management layer handles all reservation-related tasks. The Equipment Management layer tracks equipment status. The Authentication & Authorization layer controls who can access what. This separation makes the system easier to understand and easier to work with.

2. Easier Maintenance:

Changes in one area do not cause problems in areas. For example if we need to update how reservations are handled we can do it inside the Reservation Management component without affecting the Web Portal or the database. This makes updates faster and safer.

Security Advantage

The Authentication & Authorization layer stands alone which helps enforce role-based access control. Students can make their reservations but only authorized lab technicians can manage equipment. This ensures that users see what they are allowed to see. It keeps the system secure. Follows the portal's security requirements.

Performance Benefit

By keeping business logic separate from the user interface and the database the system processes requests in an organized way. For example checking if equipment is available or making a reservation happens in the layer instead of the Web Portal directly accessing the database. This reduces load and improves response times.

Why Other Architectures Were Not Selected

Microservices Architecture:

Microservices was not chosen because the portals functions are closely connected and work well together in one application. Using microservices would add complexity. It would require

network communication, which could slow things down. It could also cause problems with data consistency.

Client-Server Architecture:

Client-Server architecture offers deployment and centralized control.. It can create a single point of failure. It also struggles with scalability. These issues make it less suitable for a system, like this one that needs to manage responsibilities clearly.

Conclusion

Layered Architecture was selected because it supports separation of responsibilities enables role-based security makes maintenance easier and provides a clean organized structure. It fits the needs of the Smart Lab Equipment & Slot Reservation Portal perfectly.